

WEEK 02-STATUS

Planned Work

- Start the basic implementation of the microcontroller simulator.
- Divide the work into CPU, Memory & Stack, Instructions, and UI.
- Implement and test the basic instruction execution flow.
- Connect all four modules into one working simulator.

Completed Work

Venkateshwara U D – Worked on the CPU foundation, including registers, PC, flags, and FETCH-DECODE-EXECUTE flow.

Zainaba Fidha K N – Worked on Program Memory and basic Stack implementation.

Yathiksha U S – Worked on the 8 required instructions, their operations, and test cases.

Raaif Abdul Haamid – Worked on the basic UI with Load, Reset, Step, and Run controls.

Team Integration

The team connected the modules so that the CPU can fetch instructions from memory, decode and execute them, update the CPU state, and display the execution through the UI.

Work Carried Forward to Week 3

- Complete remaining instruction implementation and testing.
- Improve CPU-Memory-Instruction-UI integration.
- Test the complete execution flow with sample programs.